Hall Ticket Number:

MR-22

Code: B425CM5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2025 / January 2026

Object Oriented Programming through Java

(Computer Science and Engineering (AI and ML))

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 60

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	What is inheritance in Java?	1	1	1	1
1. b)	What is garbage collection in Java?	1	2	1	3
1. c)	Define a package in Java.	1	1	2	2
1. d)	What is the CLASSPATH variable used for?	1	2	2	2
1. e)	What is autoboxing in Java?	1	1	3	3
1. f)	What is the use of 'finally' block in exception handling?	1	2	3	3
1. g)	List types of events in Java.	1	1	4	1
1. h)	State the purpose of the 'Delegation Event Model'.	1	2	4	2
1. i)	List two limitations of AWT.	1	1	5	2
1. j)	Define the life cycle of an applet.	1	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Write a Java program to demonstrate exception handling using try-catch blocks.	5	1	1	2
2. b)	Discuss the concept of recursion in Java with a suitable example program.	5	2	1	3
	OR				
2. c)	Explain how inheritance works in Java with a program example.	5	1	1	2
2. d)	Write a Java program to create and manipulate an array of objects.	5	1	1	4
3. a)	Write a Java program to create and use an interface.	5	1	2	3
3. b)	Explain the costs and benefits of inheritance with examples.	5	3	2	2
	OR				
3. c)	Write a Java program to demonstrate the use of the 'final' keyword with inheritance.	5	1	2	4
3. d)	Describe how abstract classes differ from interfaces with an example.	5	2	2	3
4. a)	Explain the life cycle of a thread with examples.	5	3	3	2
4. b)	Write a Java program to demonstrate the use of generics with collections.	5	4	3	4
	OR				
4. c)	Describe the exception hierarchy in Java with examples.	5	2	3	3
4. d)	Develop a Java program to read and write strings using java.io.	5	3	3	4
5. a)	Discuss the different types of layout managers in Java with examples.	5	2	4	4
5. b)	Write a Java program to create a user interface with labels, buttons, and text components.	5	3	4	4
	OR				
5. c)	Explain the use of the 'CardLayout' manager with an example program.	5	3	4	2
5. d)	Explain the MVC architecture in Swing and how it addresses the limitations of AWT.	5	2	4	2

P.T.O.

6. a)	Discuss the architecture and components of Swing with examples.	5	4	5	4
6. b)	Write a Java program to create a user interface with JLabels, JTextFields, and JButtons.	5	3	5	4
	OR				
6. c)	Write a Java program to create a tabbed pane and add tabs with different components in each tab.	5	4	5	4
6. d)	Explain the life cycle of an applet with an example program that demonstrates each phase.	5	2	5	4

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome